//TCP Server

```
#include<stdlib.h>
#include<stdio.h>
#include<sys/types.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>
#include<string.h>
#include <arpa/inet.h>

int main()
{
	char buff[512];
	int SocketFD;
	struct sockaddr_in stSockAddr, cli_addr;
	SocketFD=socket(AF_INET,SOCK_STREAM,0);
	if(SocketFD<0)
	{
		perror("can not creat socket");
		exit(EXIT_FAILURE);
	}
	bzero(&stSockAddr,sizeof(stSockAddr));
	stSockAddr.sin_family=AF_INET;
	stSockAddr.sin_port=htons(1109);
	stSockAddr.sin_addr.s_addr=INADDR_ANY;
	if(bind(SocketFD,(struct sockaddr*)&stSockAddr,sizeof(stSockAddr))<0)
	{
		perror("can not bind");
		close(SocketFD);
		exit(EXIT_FAILURE);
	}
	int a=sizeof(cli_addr);
	listen(SocketFD,2);

	int connectFD=accept(SocketFD,(struct sockaddr*)&cli_addr,&a);
	if(connectFD<0)
	{
		perror("can not connect");
		close(SocketFD);
		exit(EXIT_FAILURE);
	}

	while(1)
	{
		printf("\n Receiving data from client.....");
		read(connectFD,buff,512);

		if(strcmp(buff,"x")==0)
			break;
		printf("\n data is:%s",buff);
```

```
                printf("\nenter data to be sent");
                //scanf("%s",buff);
                gets(buff);
                write(connectFD,buff,512);
        }
        close(connectFD);
        printf("\n");
        return EXIT_SUCCESS;
}
```

//TCP Client

```
#include<stdlib.h>
#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<unistd.h>
#include<string.h>
#include <arpa/inet.h>

int main()
{
        char buff[512];
        int SocketFD;
        struct sockaddr_in stSockAddr;
        SocketFD=socket(AF_INET,SOCK_STREAM,0);
        if(SocketFD<0)
        {
                perror("can not creat socket");
                exit(EXIT_FAILURE);
        }
        bzero(&stSockAddr,sizeof(stSockAddr));
        stSockAddr.sin_family=AF_INET;
        stSockAddr.sin_port=htons(1109);
        stSockAddr.sin_addr.s_addr=inet_addr("127.0.0.1");
        int xyz=connect(SocketFD,(struct sockaddr*)&stSockAddr,sizeof(stSockAddr));
        if(xyz<0)
        {
                perror("can not connect");
                close(SocketFD);
                exit(EXIT_FAILURE);
        }
        while(1)
        {
                printf("\nenter data to be sent");
                //scanf("%s",buff);
                gets(buff);
```

```
                write(SocketFD,buff,512);

                printf("\n Receiving data from Server.....");
                read(SocketFD,buff,512);

                if(strcmp(buff,"x")==0)
                        break;
                printf("\n data is:%s",buff);
        }
        printf("\n");
        close(SocketFD);
}
```

Output:



